

(SAHAI-Multimodal Depression Detection)

Submitted in partial fulfillment of the requirements

of the degree of

Bachelor of Engineering

By

- 1.Vinit Sushil Navlakha
- 2.Shwet Banktesh Tiwari
- 3.Mahek Yusuf Shah

(Roll No. 32,53,63)

Supervisor (s):

(Prof. Ichhanshu Jaiswal)



Department of Computer Science & Engineering (Data Science)

Vidyavardhini's College of Engineering & Technology



UNIVERSITY OF MUMBAI

2025-2026

CERTIFICATE

This is to certify that the project entitled “**SAHAI-Multimodal Depression Detection**” is a bonafide work of “**Vinit Sushil Navlakha,Shwet Banktesh Tiwari,Mahek Yusuf Shah**” (32,53,63) submitted to the University of Mumbai in partial fulfillment of the requirement for the award of the degree of **Bachelor of Engineering** in “**Computer Science & Engineering (Data Science)**”.

(Prof.Ichhanshu Jaiswal)
Supervisor/Guide

(Dr. Yogesh Pingle)
Project Coordinator

(Dr.Yogita Shelar)
Co-Supervisor/Co-Guide

(Dr.Satish Salunkhe)
Head of Department

(Dr.Rakesh Himte)
Principal

Project Report Approval for B. E.

This project report entitled (***SAHAI-Multimodal Depression Detection***)
by (***Vinit Sushil Navlakha, Shwet Banktesh Tiwari, Mahek Yusuf
Shah***) is approved for the degree of ***Bachelor of Engineering in
Computer Science & Engineering (Data Science)***.

Examiners

1. _____

2. _____

Date:

Place:

DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

Vinit Sushil Navlakha

(Signature)

Shwet Banktesh Tiwari

(Signature)

Mahek Yusuf Shah

(Signature)

Date: / /

Place:

ACKNOWLEDGEMENT

We wish to express heartfelt gratitude, and thanks to the guide Prof.Ichhanshu Jaiswal for his generous help, excellent guidance, lucid suggestions, and encouragement throughout our research study. His concrete directions and critical views have greatly helped us in successful completion of this work.

We take privilege to express our sincere thanks to Dr. Rakesh Himte, Principal, Mumbai University for providing right direction to achieve the success in various stages of the research journey.

We take this opportunity to thank the Management, Principal, HoD of the Department of Computer Science & Engineering (Data Science) of Vidyavardhini's College of Engineering & Technology, Vasai Rd. (W), District - Palghar for providing the support needed.

We are also thankful to all those who helped directly or indirectly in the completion of this work.

Vinit Sushil Navlakha
Shwet Banktesh Tiwari
Mahek Yusuf Shah

Date:

List of Figures

Figure No.	Description	Page No.
Fig.4.1	Proposed General Architecture	11
Fig.4.2	Proposed Architecture Framework	12
Fig.6.1	Wellness Tools Dashboard	16
Fig.6.2	Result And Reflection Screen	17
Fig.6.3	Multimodal Input & Analysis Interface	18
Fig.6.4	Conversational AI Interface	19
Fig. 6.5	Architecture-Wise Performance Comparison	22
Fig.6.6	Modality-Wise Performance Comparison	23

List of Tables

Table No.	Description	Page No.
Table 2.21	Survey of Existing Systems	7
Table 2.3.1	Architecture-Wise Performance Comparison	7
Table 5.1.1	Dataset Description and Characteristics	14
Table.5.2.2	Simulation Environment and Hyperparameter Settings	15
Table 6.1	Architecture Wise Performance Comparision	22
Table 6.2	Values and Parameters	22

TABLE OF CONTENTS

Sr. No.	Table Of Contents	Page No.
	CERTIFICATE	ii
	PROJECT REPORT APPROVAL	iii
	DECLARATION	iv
	ACKNOWLEDGEMENT	v
	LIST OF FIGURES	vi
	LIST OF TABLES	viii
	CONTENTS	ix
	ABSTRACT	x
1.	INTRODUCTION	1
	1.1 Overview	
	1.2 Historical Background	
	1.3 Motivation	2
	1.4 Proposed Approach	
	1.5 Additional Scope & Extensions	
2.	Review of Literature	3
	2.1 Literature Survey	
	2.2 Analysis	4
	2.3 Comparative Analysis	6
	2.4 Analysis of Existing Systems	
	2.5 Comparative Study	7
3.	Research Gap, Problem Definition and Objectives	8
	3.1 Research Gap	
	3.2 Problem Statement	
	3.3 Aim & Objective	9
4.	Proposed System	10
5.	Experimental Setup	13
	5.1 Details of Dataset	

5.2 Simulation Environment and Hyperparameter Settings	14
5.3 Performance Evaluation Metrics	
6. Result and Analysis	15
7. Conclusion	23
References	24
8. ANNEXURES	26
8.1 Paper Title and Details	
8.2 Paper Presentation Certificate	
8.3 VNPS Certificate	27
8.4 Oscillation Certificate	28
8.5 Full Research Paper	30
8.6 Copyright Receipt	36
8.7 Github Link of Project	37
8.8 Picture with Guide	

Abstract

Depression has become one of the serious mental disorders in the world, affecting millions of people worldwide. It becomes difficult to diagnose this disorder early because it involves subjective diagnosis and a lack of initiative on the part of the patient to seek help from experts. In recent times, there has been a surge in the number of people posting emotional expressions on various digital platforms such as social media and audio and video communication. The proposed research study aims at developing a multimodal approach towards depression diagnosis. The model will make use of three types of information namely textual, audio, and visual information to detect depression using machine learning and deep learning algorithms. Textual information will be analyzed by NLP, while audio features can be extracted using methods such as Mel Frequency Cepstral Coefficients (MFCC). Facial expressions from images will be analyzed to determine the emotional state of a patient. Hybrid models comprising CNN, Bi-LSTM, and attention mechanisms will be used for extracting spatiotemporal information. Decision-level fusion techniques are used to combine the output of various modality outputs in order to achieve higher robustness and stability in the classification process despite any noise in the data. The performance of the model is measured using accuracy, precision, recall, and F1-score measures. The experimental results show that the proposed approach performs better than the unimodal systems in detecting depression patterns. This framework provides an effective and scalable solution that can be used for early screening and monitoring of mental health issues. The proposed framework can be implemented in areas like online mental healthcare portals, wellness programs in the workplace environment, or academic institutions. Further work could include physiological signals along with optimization for running real-time on limited resources.

Chapter 1

Introduction

1.1 Overview

The occurrence of various psychological illnesses like depression has emerged as one of the most concerning public health issues in today's world. Not only does depression impact an individual's emotional state but it further hampers his/her efficiency, decision-making ability, and general quality of life. Although there exist various clinical approaches to detect these illnesses, early identification is rare due to various reasons including social stigma, lack of knowledge, etc. In the current era, as people communicate themselves over different digital platforms, one often comes across various textual, audio, and visual cues from users about their feelings and actions. This allows one to assess any mental illness in people by means of computations. The presented project attempts to develop a multi-modal framework for detecting depressive patients based on text, audio, and image data.

1.2 Historical Background

Diagnosis of depression conventionally involves interviewing the patient and performing a psychometric assessment. Though efficient, this method is quite time-intensive, subjective, and cannot be scaled up to accommodate mass screenings. Thanks to the development of Artificial Intelligence, machine learning has been used in analyzing the state of mind. Initially, the technique used text-based sentiment analysis, employing TF-IDF approach and classical machine learning algorithms such as SVM. Though effective in their own way, these methods did not provide adequate context and robustness. Further advancements brought along deeper neural networks such as RNN, LSTM, and CNN. These approaches significantly increased the possibility of identifying more intricate patterns; however, they remained unimodal in nature, which was a limiting factor. As a result, multimodal models came about and started being employed to detect the patient's mood.

1.3 Motivation

The main driver for the implementation of this work is the need for the development of an automatic, early, and scalable depression detection framework. There are many people, including those working and studying, who are under constant stress but do not consult any health practitioner because of social stigma or lack of time. Incorporation of multiple modalities of data into the system allows it to pick up on emotional cues which are not obvious when considered in isolation. This leads to a more realistic assessment of depression patterns. Finally, there have been advances in deep learning and computing which have made it possible to develop efficient and applicable frameworks such as mobile and web-based applications.

1.4 Proposed Approach

The suggested system is based on a multimodal deep learning architecture which makes use of text, sound, and image information. These inputs are processed separately using different models: Text information is processed through NLP and deep learning models including Bi-LSTM. Sound information is processed through feature extraction methods like MFCC and neural networks. Images are processed using CNN-based models for facial expression recognition. These outputs are merged together using an ensemble fusion at decision level, ensuring the effectiveness of all modalities in the ultimate output.

1.5 Additional Scope & Extensions

The model can further be enhanced by adding other forms of data like physiological data (EEGs, heart rate). In addition, using self-supervised learning methods can decrease reliance on labelled data. Other potential developments could involve enhancing the model to enable its deployment in real-time on mobile and edge devices.

Chapter 2

Review of Literature

2.1 Literature Survey

The area of depression diagnosis has made great progress owing to the advent of technologies such as Artificial Intelligence, Machine Learning, and Deep Learning. Traditional methods for mental well-being evaluation included clinical interviews and psychological tests; however, they had their drawbacks because they were subjective, time-consuming, and difficult to scale. With an increase in digitalized information, computational methods that could detect depression were investigated using different types of data. Sentiment analysis of the written text was one of the earliest methods used in detecting depressive behaviour. User-generated content on social media sites such as Twitter and Reddit was examined for any emotional cues, and algorithms such as TF-IDF, BoW, and classic machine learning models like SVM and Naïve Bayes were used. Although these methods gave some insights, they did not take context into consideration. The use of deep learning techniques like RNN and LSTM enabled better results because these techniques helped capture sequential relationships among the textual information. The Bi-LSTM model provided even better results by considering data information in both the forward and backward direction. Another important technique used was that of the CNN model to capture local features of text data, thereby enhancing its classification accuracy. Other methods that are being considered include emotion detection from audio features. Here, speech information is analyzed to derive different features such as pitch, tone, and MFCC values. Information about emotional status can be gleaned from this information, and it allows us to detect emotions such as stress, anxiety, and depression. Similarly, emotion detection from images, especially facial expression analysis using CNN models, has also gained popularity. Current research highlights the significance of multi-modal methods that combine textual, audio, and visual information for accurate recognition. Such systems make use of the complementarity of multiple sources, thereby giving a complete picture of the emotion felt by humans. Ensemble models and attention techniques have been useful in this regard, paying attention to only the salient aspects.

2.2 Analysis

Many research papers have focused on solving Depression Detection using AI technologies. A detailed review of relevant literature is presented below.

Sr.no	Paper Title	Publishing Journal	Summary	Drawbacks
1	M. Joshi, C. Mahajan, T. Korgaonkar, N. Raul, and M. N., "Web Application Based on User Interests for Mental Well-Being," 2023.	International Journal of Innovative Research in Computer and Communication Engineering (IJIRCCCE), 2023	Developed an app for post-pandemic mental well-being using SVM with Sigmoid Kernel for recommendation and Bi-LSTM for sentiment analysis. Accuracy: 85% on 20 users for 2 months.	Could use Deep CNN to reduce preprocessing. Add guiding material for intervention activities to improve user satisfaction.
2	S. Tamhane, A. Shirao, M. Shah, and D. Patil, "Emotion Recognition using Deep CNN," 2022.	International Research Journal of Engineering and Technology (IRJET), 2022	Used DCNN for facial emotion recognition; compared ResNet50, VGG16, DCNN — DCNN performed best.	Only 3 emotions (happy, neutral, sad) considered though FER2013 has 7.
3	H. Liu, X. Chen, and X. Liu, "Application of Weight Distributing Method Combining Sentiment Dictionary and TF-IDF," 2022.	IEEE Access, 2022	Combined rule-based sentiment dictionary with ML using weight distributing for sentence vectors. Accuracy: 82.1%.	Limited to single sentences; imbalance affects classification; weaker neutral sentiment detection.
4	P. Durga and D. Godavarthi, "Deep-Sentiment: Decision-Based Recurrent Neural Network," 2023.	IEEE Transactions on Affective Computing, 2023	Used BERT-large-cased model, SGD optimizer, and D-RNN for aspect-based sentiment analysis across Twitter, Restaurant, and Laptop datasets.	Limited to text-based analysis; lacks application for visual/audio data.
5	A. Ghorbanali and M. K. Sohrabi, "Comprehensive Survey on Deep Learning-based Multimodal Sentiment Analysis," 2023.	Journal of Intelligent Systems, 2023	Compared sentiment analysis approaches; found OTMM (91.2%) and MF-BERT effective for multimodal data.	OTMM time-consuming; MF-BERT large in size; CNN pooling may delete key info.
6	T. Jiang, S. Deng, P. Wu, and H. Jiang, "Real-Time Human-Music Emotional Interaction," 2023.	IEEE International Conference on Multimedia and Expo (ICME), 2023	Proposed 3D sentiment label (P-A-D) using CNN for combining text and image sentiment.	Suggests expanding to multi-dimensional sentiment labeling for precision.
7	G. Kumar et al., "Context-Sensitive Multi-Tier Deep Learning Framework," 2023.	IEEE Access, 2023	Three-tier CSMDf framework combining CNN, 3D-CNN, and BiGRU for multimodal feature fusion.	Recommends using transfer learning and fuzzy logic ranking for enhancement.
8	L. Liu, Y. Wang, and W. Chi, "Image Recognition Technology Based on Machine Learning," 2017.	International Journal of Computer Science and Information Security (IJCSIS), 2017	Used genetic algorithm to optimize BP neural network for license plate recognition.	Low accuracy without frames; limited anti-interference ability.
9	N. Sharma, V. Jain, and A. Mishra, "Analysis of CNNs for Image Classification," 2018.	International Conference on Computational Intelligence and Data Science (ICCIDIS), 2018	Compared CNN models; AlexNet 13%, GoogleNet 68.95%, ResNet50 52.55%.	Limited dataset; larger dataset can improve model accuracy.
10	C. E. J. Li and L. Zhao, "Emotion Recognition using CNN," 2019.	IEEE Conference on Multimedia and Signal Processing (CMSP), 2019	CNN-based model for facial emotion recognition with reduced preprocessing and lighting	Overfitting issue; fear emotion dominates classification.

			variation issues.	
11	A. Jaiswal, A. K. Raju, and S. Deb, "Facial Emotion Detection using Deep Learning," 2020.	International Journal of Engineering Research & Technology (IJERT), 2020	CNN model tested on FER-2013 (70.14%) and JAFFE (98.65%) datasets for facial emotion detection.	Needs evaluation on computational efficiency and validation loss across datasets.
12	M. S. Akhtary et al., "Multi-task Learning for Multimodal Emotion and Sentiment Recognition," 2020.	IEEE Transactions on Affective Computing, 2020	Combined text, audio, and visual modalities using multi-task learning; outperformed SOTA methods.	Did not predict sentimental and emotional intensity.
13	P. Chitkara et al., "A Stem to Stern Sentiment Analysis Emotion Detection," 2022.	Springer Nature Computer Science, 2022	Reviewed emotion models, levels, and sentiment extraction challenges.	Issues: misspelled words, domain dependency, sarcasm, negation detection.
14	N. C. Dang et al., "Sentiment Analysis Based on Deep Learning: A Comparative Study," 2020.	IEEE Access, 2020	Compared DL models — CNN best for balance, RNN better reliability but slower; dataset choice impacts results.	Requires broader datasets for deeper evaluation.
15	D. Muzafer et al., "Machine Learning Algorithms for Depression Detection," 2023.	Journal of Affective Disorders Reports, 2023	Found SVM with 82% accuracy for depression detection.	Could use paraphrase-based features for richer emotion capture.
16	K. H. S. L. Kanakkahewa et al., "PoS Tag-based Attention for Feature Selection in Sentiment Analysis," 2023.	International Conference on Machine Learning and Applications (ICMLA), 2023	Proposed PoPoSBert using BERT for contextual and syntactic feature extraction.	Limited semantic depth; misses subtle linguistic nuances.
17	D. Suryadi, "Lexicon-based Sentiment Analysis on COVID-19 News Tweets," 2020.	Indonesian Journal of Electrical Engineering and Computer Science (IJEECS), 2020	Used NRC Affect Intensity Lexicon (8 emotions) for media sentiment analysis.	Should adopt automated lexicon adaptation for objectivity.
18	A. Umair et al., "Vaccine Sentiment Analysis using BERT and NBSVM," 2023.	IEEE International Conference on Artificial Intelligence and Data Engineering (AIDE), 2023	Combined BERT with Naïve Bayes + SVM for short-text sentiment classification.	Extend model to multilingual datasets.
19	N. Dhariwal et al., "Audio and Text Sentiment Analysis of Radio Broadcasts," 2023.	IEEE International Conference on Big Data Analytics (ICBDA), 2023	Used Vokaturi (audio), AssemblyAI (transcription), and VADER (text) for multimodal sentiment detection.	NVV analysis requires more robust emotion detection integration.
20	D. Chen et al., "Emotion and Sentiment Analysis for Intelligent Customer Service," 2023.	IEEE Access, 2023	Multi-task ensemble using CNN, LSTM, GRU, and MLP for customer dialogue emotion analysis.	Should validate accuracy across different domains.

Table 2.2.1 Survey of Existing Systems

2.3 Comparative Analysis

Model Architecture	Methodology	Accuracy (%)	Precision (%)	F1-Score (%)
STFS_TF-IDF_HDSM	Hybrid Deep Sequential	98.46	96.71	93.00
FANCFIS	Fuzzy Deep Learning	99.02	98.56	98.45
DWSN-MNN	Wavelet Scattering	99.95	99.95	99.95
Ensemble ML	SVM + RF + Logistic	85.82	86.01	85.81
Baseline CNN	13-Layer CNN	70.32	67.11	77.25

Table 2.3.1 Architecture-Wise Performance Comparison

2.4 Analysis of Existing Systems

Despite considerable developments, current depression detection systems have many weaknesses. Most of the models are built on datasets with single modality, limiting their effectiveness at recognizing the full spectrum of emotions. For example, text-based approaches cannot distinguish sarcasm or hidden emotions, and audio/visual methods are limited by environmental conditions. The use of inadequate and insufficiently diverse datasets can also be considered a drawback. While most papers focus on using social media content, many of them are based on small datasets collected under controlled circumstances that do not reflect real-world situations. Moreover, some of the deep learning-based models used in depression detection systems have great accuracy, but they are very demanding regarding computational power and need extensive resources for training and deploying. Furthermore, there is no link between domain knowledge and the data approach. Most of the models use only the data component and ignore psychological aspects, which might have enhanced understanding and validity.

2.5 Comparative Study

Comparison among traditional methods, machine learning methods, and deep learning approaches shows that deep learning-based models achieve much better performance in terms of accuracy and adaptiveness compared to conventional approaches. Nevertheless, there are limitations in unimodal deep learning frameworks that fail to capture various emotional cues. A multimodal approach appears more effective since it utilizes multiple sources of information. Utilization of multiple sources such as text, sound, and images can contribute to improving the performance and efficiency of the model, particularly when some sources may have limitations. The proposed model exploits the advantages of multimodality and hybrid deep learning architecture together with ensemble fusion methodologies.

Chapter 3

Research Gap, Problem Definition and Objectives

3.1 Research Gap

Despite improvements in depression detection based on machine learning and deep learning approaches, there are several gaps in existing literature. Most current studies depend on single-modality data, limiting the scope for detecting the complex nature of emotions. Models that use text-based input often suffer from challenges posed by sarcastic language, non-formal writing, and contextual ambiguity. In addition, existing research studies often employ small-scale or restricted datasets. Inconsistency and accuracy of models can be compromised due to limitations in sample size. Combining different sources of data like text, speech, and image is an area of less emphasis, especially regarding feature integration and model efficiency. There is also a need for scalable systems with high efficiency to ensure that they can function effectively in a dynamic environment. Deep learning models have a relatively high computational cost, which makes their implementation difficult. Besides, other factors like privacy concerns and model interpretability should receive more attention.

3.2 Problem Statement

The main aim of the proposed project is to create an efficient and effective multi-modal system for depression detection that can incorporate text, speech, and video inputs using modern machine learning approaches. The problem with the currently existing technologies is that they are unable to efficiently use multiple modalities. There is also the issue of creating an efficient system that can work on unstructured and noisy data without compromising on the privacy of users.

3.3 Aim & Objective

Aim:

The aim of this project is to design and implement a **multimodal depression detection system** that leverages deep learning techniques to analyze text, audio, and visual data for early identification of depressive tendencies.

Objectives:

- 1. To develop a system capable of analyzing social media text using NLP techniques.**
- 2. To extract emotional features from audio signals using signal processing methods.**
- 3. To analyze facial expressions using deep learning-based image processing techniques.**
- 4. To integrate multiple modalities using ensemble fusion methods.**

Chapter 4

Proposed System

The system under discussion is a multichannel depression detection algorithm that makes use of hybrid deep learning techniques in order to process different types of modality including text, audio, and images in order to achieve precise results. The proposed system will be comprised of three major parts: text extraction part, audio extraction part, and image extraction part. In all three cases, different kinds of features will be extracted independently and then merged in order to produce the final results. For the processing of textual data, different natural language processing techniques along with deep learning models like bi-directional LSTM networks will be used. For audio data processing, mel-frequency cepstral coefficients will be computed first followed by deep learning-based emotional classification. The results generated by each of the above modules are merged together through a decision-level ensemble fusion approach. The use of this kind of approach ensures that there is robustness in the prediction system and it can tolerate some noise or missing information in any of the modalities. The overall prediction model should be flexible enough to enable integration into other applications such as mental well-being apps, virtual assistants, and health care services.

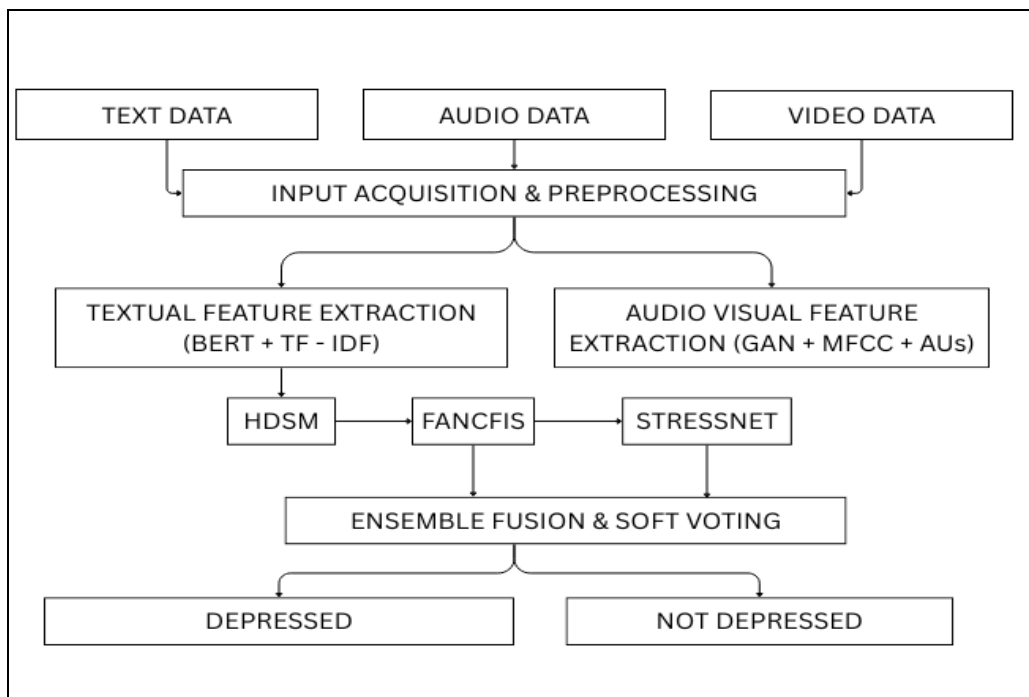


Fig.4.1 Proposed General Architecture

The architecture of the multimodal framework has been structured as a module system with the aim of integrating different data types towards its improved diagnostic precision. The multimodal framework is conducted through four main stages: Post Set Identification & Preprocessing, Textual Sentiments Analysis, Audiovisual Feature Extraction and Ensemble-Based Decision Fusion.

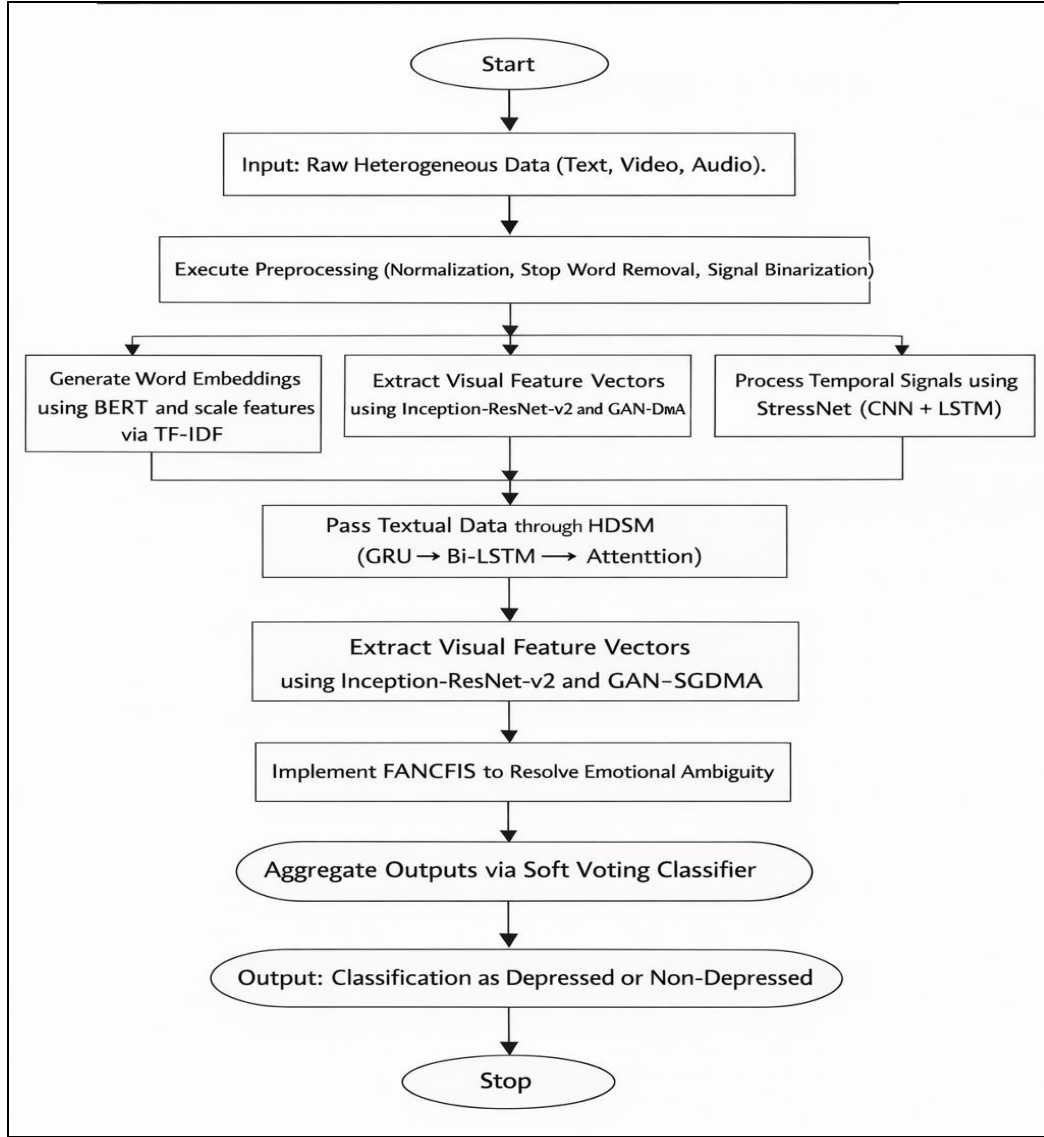


Fig.4.2 Proposed Architecture Framework

Post-Set Identification and Preprocessing Initially, a post-set $\{D_i, C_i\}$ is identified, where $I = 1, 2 \dots n$ represents the dataset size, and each entry D_i is labelled with a classification $C_i \in \{\text{Yes, no}\}$ to denote the stress or depressive state of the user.

Text Preprocessing: Standardization of the input is carried out using normalization, stop words and stemming to remove unnecessary characters and stop words not conveying meaningful information. Word embedding is done through the BERT model. BERT is short for Bidirectional Encoding of Representations from Transformers.

Image Preprocessing: Non-textual information is converted from a raw recording into a digital signal in one dimension. Image processing involves conversion to a grayscale representation to simplify processing, as well as segmentation to align features.

Textual Analysis Module (HDSM and FANCFIS): This approach follows a hierarchical one in analyzing the linguistic pattern that will help in identifying whether the sentiments can be classified as suicidal or non-suicidal.

Hybrid Deep Sequential Model (HDSM): Features are scaled using TF-IDF to quantify the importance of terms. It is then followed by a three-layer sequential architecture: Layer 1-GRU (Gated Recurrent Unit) by means of unidirectional manner to catch temporal dependencies, and Layer 2-Bi-LSTM network to exploit both past and future contexts of the input sequence. Finally, an attention mechanism was incorporated into it that allows selecting the most informative linguistic segments.

Fuzzy Logic Integration: To deal with the natural vagueness found in human emotive expressions, FANCFIS (Fast Adaptive Neuro Complex Fuzzy Inference System) is implemented. FANCFIS uses features such as the HMLSTM (Hierarchical Multi-scale) approach to learn the lower - level as well as the higher-level features in the sentiment.

Audiovisual and Image Modules (GAN-SGDMA and Stress Net): The behavioural markers are inferred based on non-verbal features to generate a richer construct for mental state assessment.

Visual Feature Extraction: Seven types of facial expression are present, corresponding to various types of emotion, which are extracted using GAN (Generative Adversarial Networks) [3]. In addition, feature vectors are autonomously extracted using the Inception Resnet V2 Model from the video frame and optimal weights are improved by SGDMA (Stochastic Gradient Descent with Momentum Algorithm)

Audio and Temporal Dynamics: The architecture of Stress Net, incorporating 2D-CNN and LSTM, was used to address the temporal dynamics of behavioural signals. Acoustic features were filtered through MFCC (Mel Frequency Cepstral Coefficients) to detect discrepancies in audio and visual streams.

Ensemble Fusion Layer The last stage in the classification is implemented using the modality outcomes in decision level fusion.

1. **Ensemble Learning:** Several base-level classifiers, such as SVM (Support Vector Machine) and Random Forest, are combined in a single model to enhance accuracy over individual models.
2. **Soft Voting Classifier:** In the design of the system, the soft voting classifier is used for aggregating the predictions of the different parts of the system that include the textual part, the visual part, and the audio part on the basis of the cumulative probability scores generated by the different parts of the system. In this way, the system can handle noisy information while relying on high-confidence predictions for diagnosis.

Chapter 5

Experimental Setup

5.1 Details of Dataset

Dataset	Modality	No. of Samples	Classes
SST-2	Text	67,349	Positive / Negative
Sentiment140	Text	1.6 Million	Positive / Negative
Amazon Electronics Reviews	Text	1.68 Million	Positive / Negative
GMC Dataset	EEG	30	Depressed / Healthy
MODMA	Multimodal (EEG, Audio, Video)	53	Depressed / Healthy

Table 5.1.1 Dataset Description and Characteristics

Benchmark Sentiment Corpora: For phrase-level binary sentiment analysis, the Stanford Sentiment Treebank2 (SST2) dataset was used as a benchmark. For large-scale testing, the Sentiment140 dataset, having 1.6 million labelled tweets and the Electronics dataset consisting of 1.68 million Amazon product reviews were used.

Clinical EEG Datasets: GMC provided EEG record data from 30 patients, consisting of 15 healthy and 15 depressed patients, while the Multi-modal Open Dataset for Mental-disorder Analysis (MODMA) had 53 participants in total.

Preprocessing and Partitioning: Standardized pre-processed steps such as normalization, stop word removal, and stemming were taken to remove noise from the input text [1]. The data is split into a ratio of 80% and 20% for the purposes of training and validation, respectively [8]. Alternatively, a fixed split of 75% for training and 25% for testing has also been taken [1].

5.2 Simulation Environment and Hyperparameter Settings

Parameter	Value
Programming Language	Python 3.7
Deep Learning Libraries	TensorFlow, Keras
Signal Processing Tool	MATLAB 2021a
Optimizer	Adam
Learning Rate	0.0001
Epochs	40 – 200
Early Stopping	Patience = 10
Train/Test Split	80/20 and 75/25
Hardware	16 GB RAM, 14-Core CPU

Table.5.2.2 Simulation Environment and Hyperparameter Settings

5.3 Performance Evaluation Metrics

Standard indicators were used to quantify the effectiveness of models like Accuracy, Precision, Recall, F1-score, and the Kappa coefficient. Accuracy was evaluated as the ratio of correct predictions to total predictions. Precision and Recall were used for measuring the capability of the classifier to model psychological health and correctly find the positive instances. The Kappa value was estimated to find the agreement between the predicted labels and ground truth, especially for the classification of MDD Architecture.

Chapter 6

Result and Analysis

The experimental outcomes show that the proposed multimodal system performs better than unimodal systems in recognizing depression. The combination of various sources of data increases the accuracy of the system. The performance of the model is impressive when measured by different parameters. It shows how efficient the model is at recognizing emotions.

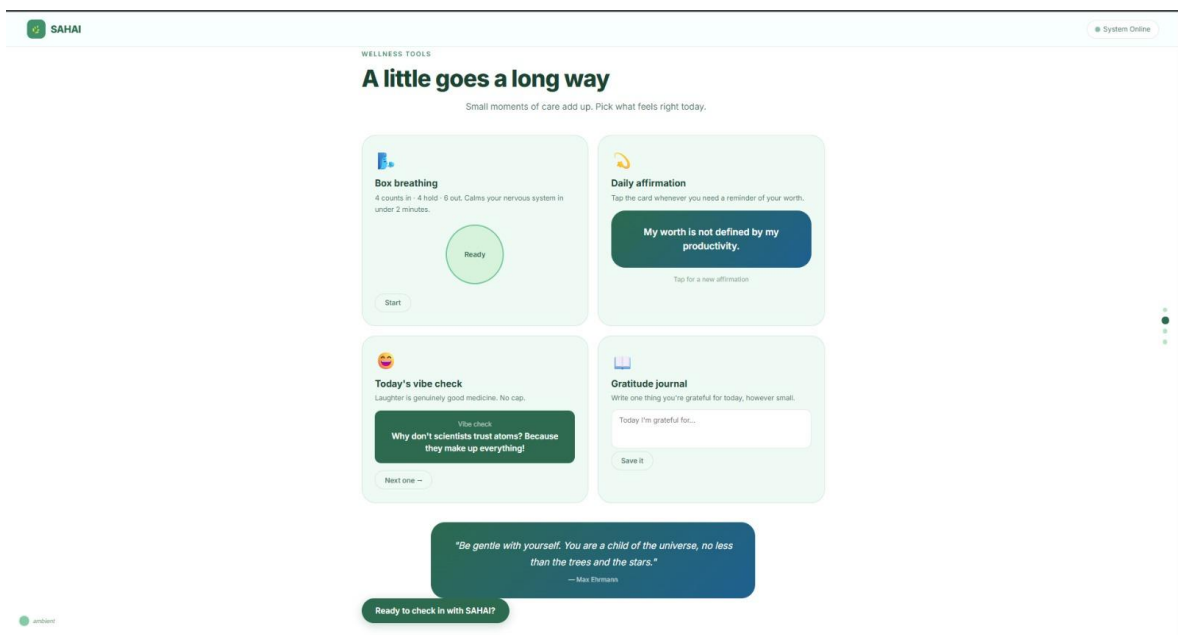


Fig.6.1 Wellness Tools Dashboard

This screen represents the wellness tools dashboard of the SAHAI system, designed to promote user engagement and emotional awareness before initiating a full depression analysis. The interface provides multiple lightweight mental health activities such as box breathing, daily affirmations, vibe checks, and gratitude journaling. These features are intentionally designed to encourage users to reflect on their emotional state in a non-intrusive and supportive manner. From a system perspective, this module acts as a pre-interaction layer, helping to collect subtle behavioural signals and user input patterns. The minimalistic UI design with soft colour tones enhances usability and reduces cognitive load, which is important for users experiencing stress or emotional discomfort.

Additionally, the dashboard serves as a gateway to the main multimodal analysis system, allowing users to transition naturally from self-care activities to structured mental health evaluation. This improves user retention and ensures that data collection begins in a comfortable and user-friendly environment.

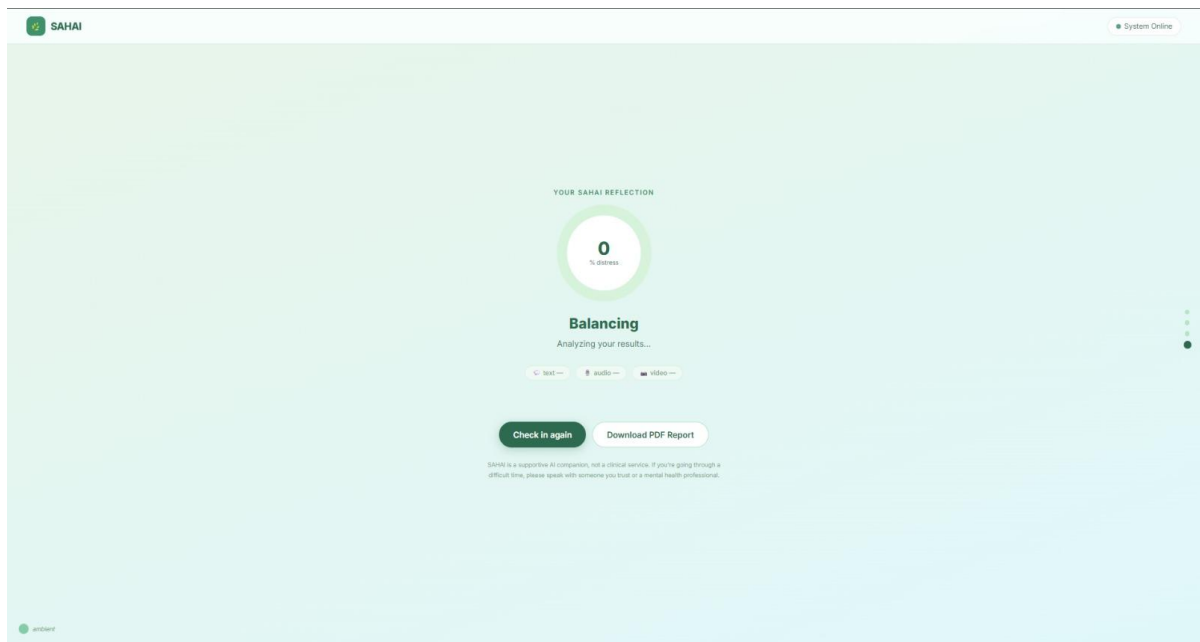


Fig.6.2 Result and Reflection Screen

This screen displays the final output of the multimodal depression detection system, presenting the user’s emotional assessment in a simplified and interpretable format. The circular indicator represents the overall depression score, while the textual label (e.g., “Balancing”) provides a qualitative interpretation of the result. The system integrates predictions from multiple modalities—text, audio, and video—using an ensemble fusion approach, and this screen reflects the aggregated outcome. The design focuses on clarity and avoids overwhelming the user with excessive technical details. Additional features such as “Check in again” and “Download PDF Report” provide functionality for continuous monitoring and record keeping. From a technical standpoint, this module plays a crucial role in model output visualization and user feedback, ensuring that complex machine learning results are translated into meaningful insights for the user.

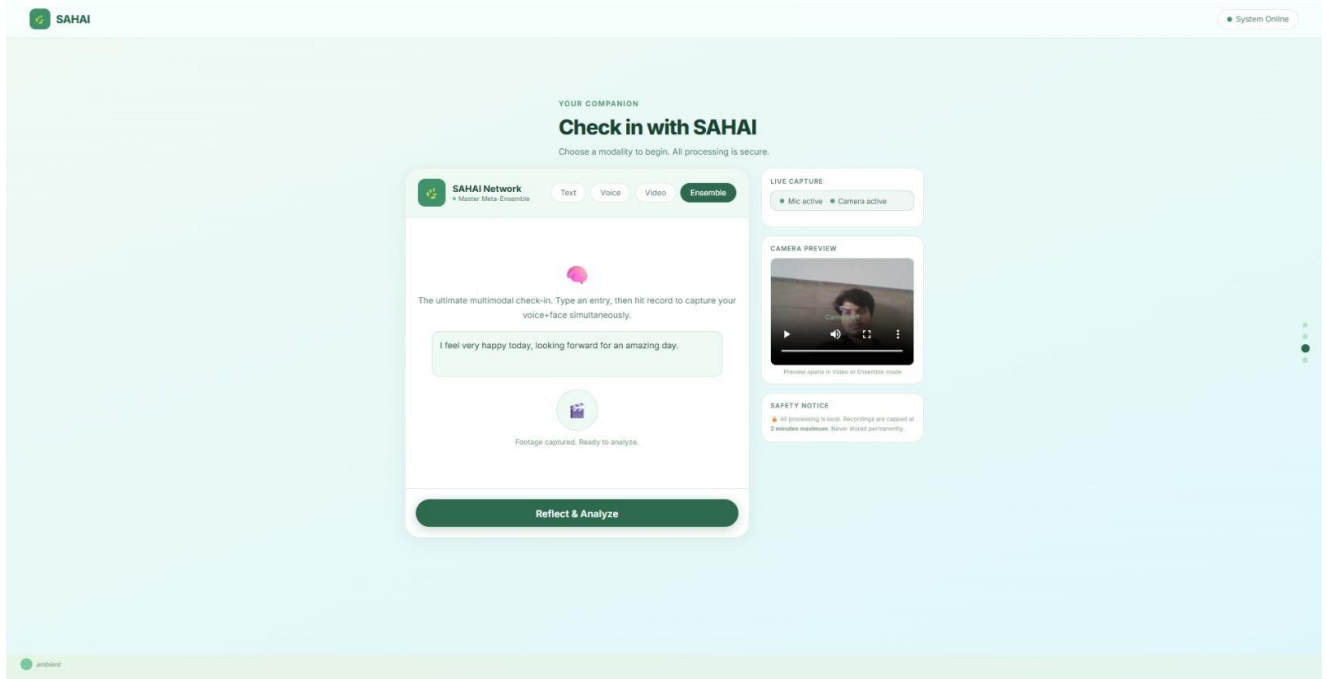


Fig.6.3. Multimodal Input &Analysis Interface

This interface is the core interaction module where users perform a multimodal check-in using text, voice, and video inputs. The system allows selection between different modalities or a combined “ensemble” mode for comprehensive analysis. The central panel captures textual input, while the side panel handles live audio and video capture, enabling real-time data acquisition. Features such as microphone and camera activation indicate active data streams, which are processed using NLP models, speech emotion recognition algorithms, and facial expression analysis techniques. This module demonstrates the practical implementation of cross-modal data fusion, where heterogeneous inputs are collected simultaneously. The structured layout ensures that users can easily interact with the system without technical complexity, while the backend processes data using deep learning models such as CNNs and Bi-LSTMs.

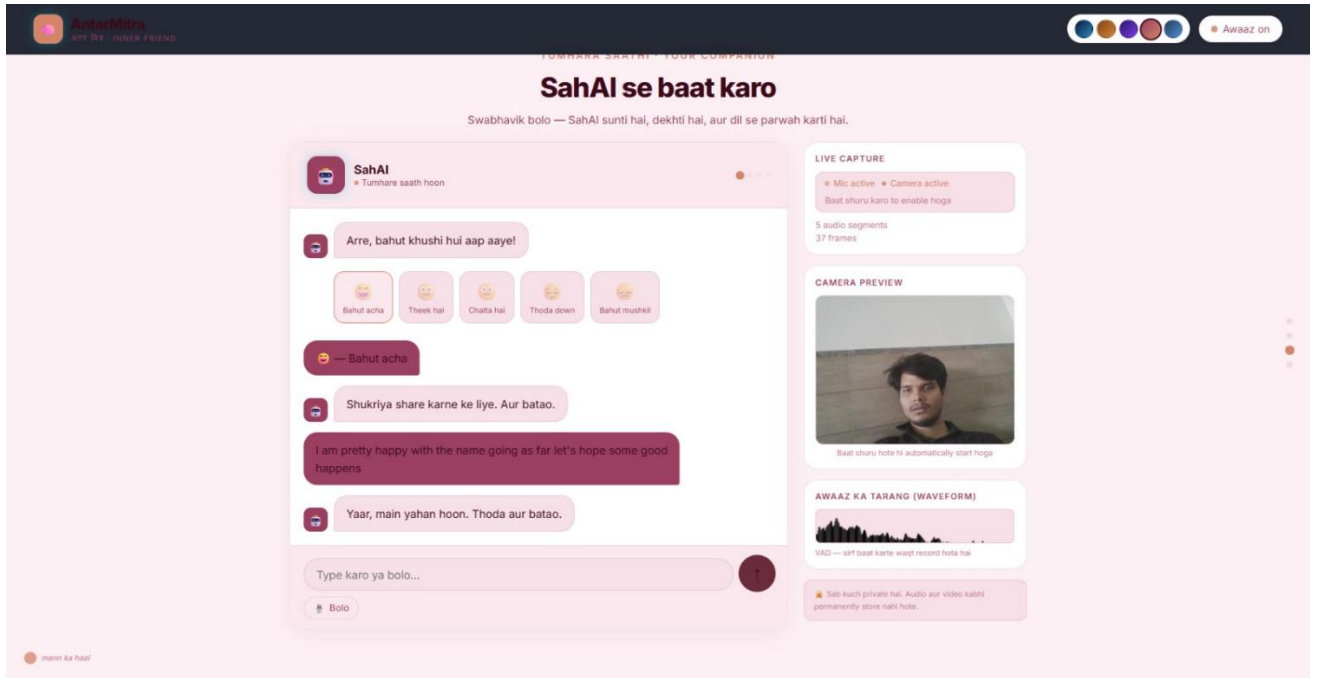


Fig.6.4 Conversational AI Interface

This screen illustrates the conversational interface of the SAHAI system, designed to simulate a supportive and empathetic dialogue with the user. The chatbot interacts in a natural, human-like manner, encouraging users to express their thoughts and emotions freely. The system supports multilingual interaction (including Hindi), making it more accessible and culturally relevant. Alongside the chat, real-time audio and video capture modules are active, enabling continuous collection of multimodal data during conversation. From a technical perspective, this module integrates natural language understanding with emotional context analysis, allowing the system to respond appropriately while simultaneously feeding user responses into the depression detection model. This conversational approach reduces user hesitation and creates a more comfortable, stigma-free environment for mental health assessment.

To evaluate the performance of the proposed framework, a set of experiments were conducted on a system with a size 16 GB RAM and a CPU of 14 cores. Simulations were conducted by using the Python environment for the Python 3.7 programming language with the addition of the Keras and TensorFlow libraries. In the case of signal processing within the clinical environment, simulation was conducted using the MATLAB 2021a programming environment for the Wavelet Scattering Transform (WST).

Dataset Description and Preparation

The datasets used for evaluation are summarized in Table II. Ensure the generalizability of the deep learning model, both the benchmark linguistic databases and the clinical electrophysiological data were used:

- 1) Benchmark Sentiment Corpora: For phrase-level binary sentiment analysis, the Stanford Sentiment Treebank2 (SST2) dataset was used as a benchmark. For large-scale testing, the Sentiment140 dataset, having 1.6 million labelled tweets and the Electronics dataset consisting of 1.68 million Amazon product reviews were used [1].
- 2) Clinical EEG Datasets: GMC provided EEG record data from 30 patients, consisting of 15 healthy and 15 depressed patients, while the Multi-modal Open Dataset for Mental-disorder Analysis (MODMA) had 53 participants in total. [4]
- 3) Preprocessing and Partitioning: Standardized preprocessed steps such as normalization, stop word removal, and stemming were taken to remove noise from the input text [1]. The data is split into a ratio of 80% and 20% for the purposes of training and validation, respectively [8]. Alternatively, a fixed split of 75% for training and 25% for testing has also been taken [1].

Simulation Parameters and Optimization

The technical parameters were constantly fine-tuned while trying to optimize the combination of precision and simplicity [8].

Optimization Layer:

In this layer, Adam optimizer was used for controlling the rate of convergence, and the initial learning rate was assigned 0.0001 for better stability in convergence performance [1] [2].

Training Schedule:

When training the given model, different epochs were established, ranging from 40 to 200 [1]. As a preventive measure against overfitting of the data within the training period, the inclusion of early stopping with a patience of 10 epochs has been integrated into the model [6].

Feature Extraction Settings:

An optimizer was applied to train a network by fine-tuning the inception resnet v2 vision module of 2000 steps with a learning rate of 0.0001 [3]. The scale of WST invariance was chosen to be half the signal's length [4].

Performance Evaluation Metrics

Standard indicators were used to quantify the effectiveness of models like Accuracy, Precision, Recall, F1-score, and the Kappa coefficient. Accuracy was evaluated as the ratio of correct predictions to total predictions. Precision and Recall were used for measuring the capability of the classifier to model psychological health and correctly find the positive instances. The Kappa value was estimated to find the agreement between the predicted labels and ground truth, especially for the classification of MDD Architecture-wise and modality-wise and performance comparisons.

Model Architecture	Methodology	Accuracy (%)	Precision (%)	F1-Score (%)
STFS_TF-IDF_HDSM	Hybrid Deep Sequential	98.46	96.71	93.00
FANCFIS	Fuzzy Deep Learning	99.02	98.56	98.45
DWSN-MNN	Wavelet Scattering	99.95	99.95	99.95
Ensemble ML	SVM + RF + Logistic	85.82	86.01	85.81
Baseline CNN	13-Layer CNN	70.32	67.11	77.25

Table 6.1 Architecture Wise Performance Comparison

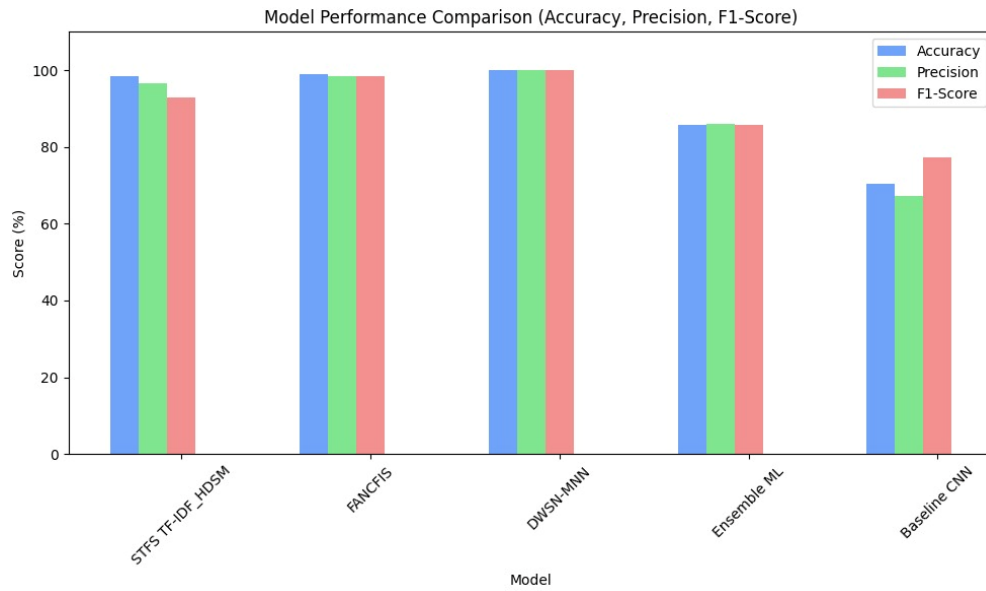


Fig. 6.5. Architecture-Wise Performance Comparison

Quantitative Results and Discussion

Parameter	Value
Optimizer	Adam
Learning Rate	0.0001
Epochs	40–200
Early Stopping	Patience = 10
Train/Test Split	80/20, 75/25

Table 6.2. Values and Parameters

The experimental findings show that multimodal ensemble frameworks achieve significantly higher accuracy than standalone unimodal architectures. As demonstrated in Table 6.2, the integration of heterogeneous features leads to superior classification stability [2] [10] [4].

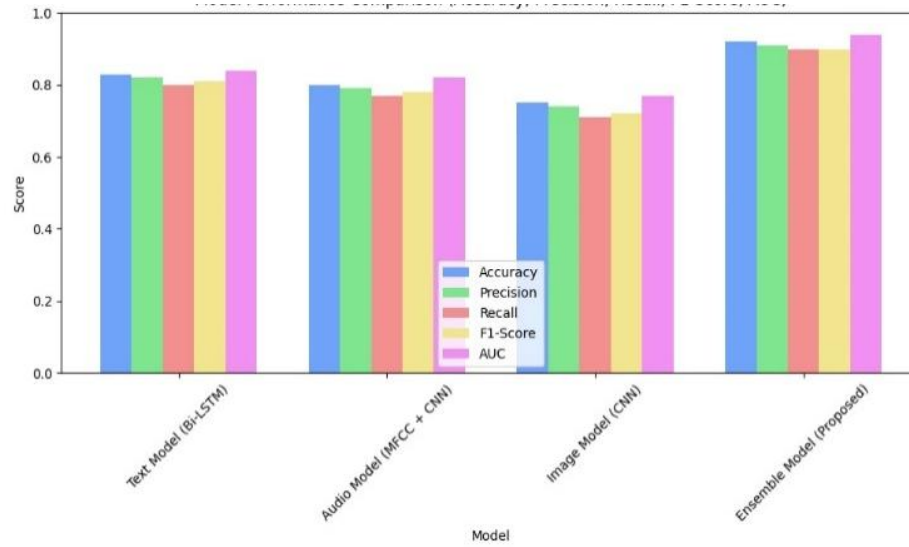


Fig.6.6 Modality-Wise Performance Comparison

It was noted that the Hybrid Deep Sequential Model (HDSM) was able to obtain the minimum loss mode of 0.3121. This emphasizes the effectiveness of the model in recognizing the emotional flow using the Bi-Long Short-Term Memory model [5]. In the analysis of the Fast Adaptive Neuro-Complex Fuzzy Inference System (FANCFIS) for the GMC data set, an effective management of the ambiguity involved in the expressions related to sentiments was observed [1]. In the analysis of the right hemisphere EEG signal compared to the left hemisphere for the GMC data set, the higher classification accuracy attained through the right hemisphere emphasizes the involvement of the brain's right hemisphere in the detection of depression [4]. Accordingly, these analysis results emphasize the effectiveness of the proposed approach involving the use of the Cross-Modal Synergy for the improvement of precision in the professional environment through the achievement of higher AUC values up to 1 for the Scattering Networks [4].

Chapter 7

Conclusion

The present research successfully proves the efficiency of the proposed framework in the context of early depression screening for working professionals. The proposed architecture successfully reduces the subjectivity and scalability limitations of traditional screening processes by utilizing heterogeneous data sources such as text data, audiovisual data, and electrophysiological data. The experimental results confirm the robustness of the cross-modal interaction effect. The Hybrid Deep Sequential Model and its components show high classification accuracy up to 98.46%, while the Deep Wavelet Scattering Networks show a peak classification accuracy of 99.95%. The results reinforce the effectiveness of the proposed framework in combining semantic scaling (TF-IDF), sequential deep learning models (GRU and Bi-LSTM with attention), fuzzy inference systems, and decision-level fusion. As illustrated in the modality-wise comparison in Table VI, the proposed ensemble model outperforms the individual unimodal models in terms of accuracy, precision, recall, and AUC values. Nonetheless, challenges are noted with respect to computational complexity, real-time applicability, and model interpretability. Future research directions should be focused on including more physiological biomarkers, as well as Explainable AI techniques for enhanced model trust from a clinical point of view. The model can be further improved with respect to the cross-attention mechanisms between modalities. In conclusion, it can be noted that the proposed framework for cross-modal depression screening has brought a new dimension with respect to objective and automated methods for evaluating mental health. The proposed system has high potential with respect to early risk detection and assistance from healthcare practitioners.

References

- [1] P. Bharath and D. Venkatalakshmi, “FANCFIS: Ensemble deep learning based feature learning with a novel fuzzy approach for sentiment analysis,” *International Journal of Information Technology*, vol. 16, pp. 1–12, doi: 10.1007/s41870-024-01882-2.I, May 2024.
- [2] A. Khan and R. Gupta, “Early depression detection using ensemble machine learning framework,” *International Journal of Information Technology*, vol. 16, pp. 1–13, doi: 10.1007/s41870-024-01981-0, Jun. 2024.
- [3] A. K. Cherian, M. Vaidhehi, M. Arshey, J. Briskilal, and S. V. Simpson, “Generative adversarial networks with stochastic gradient descent with momentum algorithm for video-based facial expression,” *International Journal of Information Technology*, doi: 10.1007/s41870-024-02043-1, Jun. 2024.
- [4] M. Mahajan *et al.*, “ECG signal classification via ensemble learning: Addressing intra and inter-patient variations,” *International Journal of Information Technology*, vol. 16, pp. 4931–4939, doi: 10.1007/s41870-024-02086-4, Aug. 2024.
- [5] F. R. Kareem and A. J. Obaid, “STFS_TF-IDF_HDSM: Semantic and textual feature scaling using TF-IDF based sentiment analysis with Hybrid Deep Sequential Model,” *International Journal of Information Technology*, vol. 17, no. 9, pp. 5751–5757, doi: 10.1007/s41870-024-02330-1, Dec. 2025.
- [6] N. Sharma, M. Sharma, J. Tailor, A. Chaudhari, D. Joshi, and U. R. Acharya, “Automated detection of depression using wavelet scattering networks,” *Medical Engineering and Physics*, vol. 124, Art. no. 104107, doi: 10.1016/j.medengphy.2024.104107, Jan. 2024.
- [7] M. Ying, X. Shao, J. Zhu, Q. Zhao, X. Li, and B. Hu, “EDT: An EEG-based attention model for feature learning and depression recognition,” *Biomedical Signal Processing and Control*, vol. 93, Art. no. 106182, doi: 10.1016/j.bspc.2024.106182, Mar. 2024.
- [8] S. A. M. Mane and A. Shinde, “StressNet: Hybrid model of LSTM and CNN for stress detection from electroencephalogram signal (EEG),” *Results in Control and Optimization*, vol. 11, Art. no. 100231, doi: 10.1016/j.rico.2023.100231, May 2023.

- [9] S. Yasin, A. Othmani, I. Raza, and S. A. Hussain, "Machine learning based approaches for clinical and non-clinical depression recognition and depression relapse prediction using audiovisual and EEG modalities: A comprehensive review," *Computers in Biology and Medicine*, vol. 159, Art. no. 106741, doi: 10.1016/j.combiomed.2023.106741, Mar. 2023.
- [10] V. Vajrobol, N. Aggarwal, U. Shukla, G. J. Saxena, S. Singh, and A. Pundir, "Explainable cross-lingual depression identification based on multi-head attention networks in Thai context," *International Journal of Information Technology*, vol. 15, doi: 10.1007/s41870-023-01512-3, Dec. 2023.
- [11] J. Cao, Y. Zhang, X. Wang, and J. Liu, "Deep learning-based multimodal depression detection using audio, visual and textual cues," *IEEE Transactions on Affective Computing*, vol. 14, no. 2, pp. 987–999, doi: 10.1109/TAFFC.2022.3145678, Apr. 2023.
- [12] S. AlHanai, M. Ghassemi, and J. Glass, "Detecting depression with audio/text sequence modeling of interviews," *Proceedings of Interspeech*, pp. 1716–1720, doi: 10.21437/Interspeech.2018-2522, Sep. 2018.
- [13] H. T. Nguyen, D. Tran, and S. Venkatesh, "Depression detection from social media text using deep neural networks," *IEEE Access*, vol. 8, pp. 102343–102354, doi: 10.1109/ACCESS.2020.2998423, Jun. 2020.
- [14] Y. Huang, J. Epps, and M. Joachim, "Depression detection from speech signals using deep convolutional neural networks," *IEEE Journal of Biomedical and Health Informatics*, vol. 24, no. 6, pp. 1733–1742, doi: 10.1109/JBHI.2019.2941517, Jun. 2020.
- [15] X. Li, L. Zhao, and G. Liu, "EEG-based depression recognition using hybrid CNN-LSTM model," *Biomedical Signal Processing and Control*, vol. 68, Art. no. 102688, doi: 10.1016/j.bspc.2021.102688, Aug. 2021.
- [16] M. S. Hossain, G. Muhammad, and N. Guizani, "Explainable AI for depression detection from multimodal data," *IEEE Network*, vol. 35, no. 4, pp. 298–305, doi: 10.1109/MNET.011.2000495, Jul. 2021.
- [17] R. Sawhney, P. Manchanda, P. Mathur, and R. Shah, "Exploring and learning suicidal ideation connotations on social media with deep learning," *Proceedings of ACL*, pp. 1–10, doi: 10.18653/v1/2020.acl-main.1, Jul. 2020.

- [18] Z. Yang, D. Yang, C. Dyer, X. He, A. Smola, and E. Hovy, “Hierarchical attention networks for document classification,” *Proceedings of NAACL*, pp. 1480–1489, doi: 10.18653/v1/N16-1174, Jun. 2016.
- [19] A. Sehgal and P. Agarwal, “Sentiment analysis using deep learning architectures: A survey,” *IEEE Access*, vol. 8, pp. 103923–103943, doi: 10.1109/ACCESS.2020.2999108, Jun. 2020.
- [20] K. Ma, H. Zhang, and Y. Wang, “Multimodal depression recognition based on attention mechanism and feature fusion,” *Information Fusion*, vol. 71, pp. 68–79, doi: 10.1016/j.inffus.2021.01.005, Jul. 2021.

Annexures

8.1 Paper Title and Details

“SAHAI-Multimodal Depression Detection System”.

Presented at the 20th INDIACom, 2026, 13th International Conference on “Computing for Sustainable Global Development” (INDIACom) and got acceptance for publication in IEEE.

**INDIACom-2026** <indiacom@bvicam.in>
to ichhanshu, mahek.s238216201, shwet.224626101, me ▾

Sat, Mar 14, 6:17 PM ☆ ☺ ↶ ⋮

Dear Prof. / Dr. / Ms. / Mr. Shwet Banktesh Tiwari,

Congratulations!

The reviews of your paper ID: 2609 entitled: **Cross-Modal Synergy: Improving Depression Screening through Heterogeneous Data Fusion** are now complete and it is our pleasure to inform you that the consensual decision of the Technical Programme Committee (TPC) is to accept the above mentioned paper for presentation in **INDIACom-2026**; Computing for Sustainable Global Development, technically sponsored by IEEE Delhi Section, scheduled to be held during 08th - 10th April, 2026 at Bharati Vidyapeeth, New Delhi (INDIA) and publication in the Soft Copy Pre - Conference Proceedings having ISBN 978-93-80544-52-6 serials.

This acceptance is subject to completion of the registration formalities of at-least one of the authors in the applicable category, the signed copy of the Copyright Transfer Form available at <https://bvicam.ac.in/indiacom/downloads/Copyright%20Transfer%20Form.pdf> and the Certificate of Originality and Confirmation to Attend the Conference available at http://www.bvicam.ac.in/indiacom/Downloads/Certificate_of_Originality_and_Confirmation_to_attend_the_Conference.pdf. Registration Form and other details can be downloaded from http://www.bvicam.ac.in/indiacom/Downloads/Registration_Form_for_INDIACom.pdf and the payment modes can be viewed at http://www.bvicam.ac.in/indiacom/Downloads/Payment_Mode.pdf.

Kindly note that Registration Fee covers processes related to Conference Organization and Hospitality. All publications (Soft Copy / IEEE) as per applicable date and pre-notified policy, are free of cost.

You are hereby invited to personally attend the Conference during 08th - 10th April, 2026 and present the paper as per the presentation schedule. **Please note that only those papers, which are**

8.2 Paper Presentation Certificate

Ref. No. - RV / ICAM / ND / 20(285) / 2026-27

10-April-2026

**Bharati Vidyapeeth's Institute of Computer Applications and Management (BVICAM), New Delhi**

**IEEE**
Advancing Technology for Humanity
Delhi Section

**IETE, Delhi Centre**

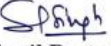
**Computer Society of India**
CSI Delhi Chapter

**ISTE, Delhi Section**

20th INDIACom; 2026 13th International Conference on
Computing for Sustainable Global Development
INDIACom - 2026
IEEE Conference ID: 70271
(08th - 10th April, 2026)

This is to certify that Prof. / Dr. / Mr. / Ms. **Shwet Banktesh Tiwari** of **Vidyavardhini's College of Engineering and Technology** has attended / contributed / presented a paper entitled **Cross-Modal Synergy: Improving Depression Screening through Heterogeneous Data Fusion** in 20th INDIACom; 2026 13th International Conference on “Computing for Sustainable Global Development”, organised by BVICAM, New Delhi, during 08th - 10th April, 2026.


(Prof. A. K. Saini)
Conference Chair


(Prof. Sunil Pratap Singh)
Convener


(Prof. M. N. Hoda)
General Chair

8.3 VNPS Certificate





8.4 Oscillation Certificate





8.5 Full Research Paper

Cross-Modal Synergy: Improving Depression Screening through Heterogeneous Data Fusion

Shwet Banktesh Tiwari
Computer Science and Engineering (Data Science)
Vidyaavardhini's College of Engineering and Technology
Mumbai, India
shwet.224626101@vcet.edu.in

Vinit Sushil Navlakha
Computer Science and Engineering (Data Science)
Vidyaavardhini's College of Engineering and Technology
Mumbai, India
vinit.224366109@vcet.edu.in

Mahek Yusuf Shah
Computer Science and Engineering (Data Science)
Vidyaavardhini's College of Engineering and Technology
Mumbai, India
mahek.s238216201@vcet.edu.in

Ichhanshu Jaiswal
Computer Science and Engineering (Data Science)
Vidyaavardhini's College of Engineering and Technology
Mumbai, India
ichhanshu.jaiswal@vcet.edu.in

Abstract— Depression is identified as a critical global health concern as well as a major cause of disability in the global arena. Enhanced levels of diagnostic accuracy for mental health evaluation are sought to be achieved with the design of a multimodal framework with integrated capabilities to incorporate textual, audiovisual, and image-based behavioral characteristics. Traditional evaluation processes are identified to face limitations in relation to the subjectivity of general physicians as well as scalability issues; nonetheless, with respect to leveraging divergent behavioral characteristics in evaluation processes, there is application of an ensemble learning architecture. Prominent levels of classification accuracy, up to 98.46%, are achieved with respect to decision-level fusion of the application of various modes of behavioral evaluation. Implementation of Early Risk Detection systems is identified as critical with respect to initiating processes of recovery to mitigate the social economic consequences of mental health conditions.

Keywords— Depression detection, Multimodal framework, Ensemble learning, Heterogeneous data fusion, Sequential deep learning, Early Risk Detection (ERD), Cross-modal synergy, Sentiment analysis.

I. INTRODUCTION

Depression is considered a debilitating emotional disorder that affects 322 million people worldwide and 4.4% of the global population, wherein it is recognized as a major cause of mental disability [6] [2] [9]. This will further increase to 35% of the world's total mortality in the year 2030 due to cardiovascular and mental health disorders that will be related to it [4]. In the case of India, over fifty million people are found to be severely affected, and the country is considered one of the most high-prevalence regions to report such cases [2]. Effective pharmacological and psychological treatments are available; however, many people do not consider approaching a professional owing to social stigma, ignorance, and/or fear of being branded in a workplace. Traditional methods of screening through self-reporting mechanisms and interviews by patients are identified as being flawed due to the involvement of physician subjectivity and the lack of scalability of the current methods. To be more specific, the methods usually involve the willingness of every individual to offer an accurate picture of the status of their mind. In the case of working professionals, such an expectation is usually flawed due to the tendency towards chronic stress or conditions [9] [6]. In relation to the above-mentioned gaps associated with current

mechanisms for achieving intelligent system-based Early Risk Detection (ERD) capabilities, the potential of social media sites and other internal communication mechanisms to serve as vast data generators is recognized. Advances made in the domain of Artificial Intelligence (AI) and Deep Learning provide an opportunity to understand these naturalistic data sources to determine mental well-being in an objective manner as follows: "Automated systems use various behavioral markers such as Mel Frequency Cepstral Coefficients (MFCCs) for acoustic features and Action Units (AUs) related to muscle movement to automatically detect various indicators of distress" [3] [9]. Preference is now given to the use of multimodal models over unimodal models because "Depressive Symptomatology has been recognized as multi-channel indicators and exhibit inter-individual variation in manifestation [7][9][4]." As per the objectives concluded for this study, the objective has been identified as follows: "Development of a scalable framework to interpret various indicators for professional individuals using ensemble-based learning paradigm" [2][4][6][9].

A. Novel Contribution

The primary objective of this research presented in this article is to identify a scalable objective framework for detecting distress. The major essential contributions of this research are identified as follows:

1. Focused User Group: It is designed for working professionals who face high stress and mental exhaustion but do not seek mental health assistance.
2. Multimodal Data Integration: It integrates data from various sources like text, audio, and images to create a better understanding of the user's emotional state.
3. Deep Learning for Pattern Detection: Deep learning techniques like Bi-LSTM and GRU with attention mechanisms are used for detecting emotional patterns in the data.
4. Ensemble Fusion for Improved Accuracy: It has been implemented at the decision level to ensure improved accuracy and minimize errors compared to single-modality systems.

II. RELATED WORK

The early studies conducted about using automated depression detection systems and techniques were heavily dependent on unimodal data sources. This means studies on linguistic analysis of texts used within social media platforms or understanding speech using acoustic analysis or facial expression using static images were some of the steps towards understanding and detecting depression early and automatically [9]. However, it was noted that these systems using unimodal approaches failed because of depressive symptoms not showing up consistently in different demographics of users. Thus, there is a major shift towards multimodal approaches [9]. Recently, many important research papers on the topic of advanced architecture for the evaluation of mental conditions were presented involving various techniques respectively:

Semantic and Textual Feature Scaling: Recently, a new method was proposed in terms of an innovative approach named STFS TF - IDF HDSM by Kareem and Obaid 2025, in which TF - IDF was employed for scaling features. It demonstrated better precision of 96.71% for identifying suicidal intention from clinical notes by employing HDSM that consisted of GRU and Bi-LSTM.

Ensemble Frameworks for Early Risk Detection (ERD): A new ensemble machine learning classifier has been advocated in the literature by Khan and Gupta (2024) to identify depression from social media sentiments by employing ensemble classifiers including Support Vectors Machines (SVM), Random Forest (RF), and Logistic Regression. It demonstrated how ensemble classifiers outperform individual classifiers by employing dominant class features and complex mental state descriptions [2].

Fuzzy Logic in Emotion Analysis: To deal with the ambiguity or impreciseness accompanying human emotional state representation, a FANCFIS (Fast Adaptive Neuro-Complex Fuzzy Inference System) method has been propounded by Potu and Venkatalakshmi (2024) where an ensemble method of deep learning technique has been adopted to incorporate Hierarchical Multi-scale LSTM (HMLSTM) along with ConvNeXt to attain an accuracy of 99.02% [1].

Physiological Signal Classification: Mahajan et al. in 2024 studied the efficiency of ensemble learning in electrocardiogram signal classification by applying a soft voting method in predicting signals. The method consists of multiple models that are combined in a single model through a voting system [4].

Audiovisual Emotion Recognition: Cherian et al. (2024) employed a framework of Generative Adversarial Networks (GAN), along with the Stochastic Gradient Descent with Momentum Algorithm (SGDMA). Furthermore, feature vector formation from the audio and video signals has been enhanced by the pre-trained Inception - ResNet - v2 model to ascertain the detection of emotion. Several gaps and challenges are identified based on literature as far as research is concerned. To start with, it is noted that there is underutilization of some of the existing deep learning models given their inability to exploit multiple frequencies or bands simultaneously [7]. Secondly, there is this challenge of "curse of dimensionality," which is likely to occur during signal-based classification tasks given its

likelihood of getting irrelevant features and thereby compromising the precision of target objects [10]. On top of this, there is a noted lack of flexibility with regards to professionals with careers who are experiencing career-based stress and are likely to conceal symptoms under zero-stigmata of work [2] [9]. Most of these works are noted to have overlooked the chances of using decision-level based on late or early fusion. A comparative summary of existing systems, including advantages and limitations, is presented in Table I. As a summary of all the reviewed literature, it can be concluded that even though there is a considerable advancement in unimodal and early fusion models, they show limitations related to subjectivity and complexity. Therefore, it is recognized that multimodal approaches have a better potential to effectively manage the complexity associated with depression. Nonetheless, the need for effectively developing fusion systems with accuracy and computational efficiencies is a challenge. Furthermore, the necessity to have a robust system to utilize Cross Modal Synergy towards ensuring a better level of consistency in the process of diagnosing professional individuals is clearly recognized.

III. PROPOSED METHODOLOGY

The architecture of the multimodal framework has been structured as a module system with the aim of integrating different data types towards its improved diagnostic precision. The multimodal framework is conducted through four main stages: Post Set Identification & Preprocessing, Textual Sentiments Analysis, Audiovisual Feature Extraction and Ensemble-Based Decision Fusion. The flow diagram of the integrated framework is as shown in Fig.1 while the integrated framework is detailed in Fig. 2.

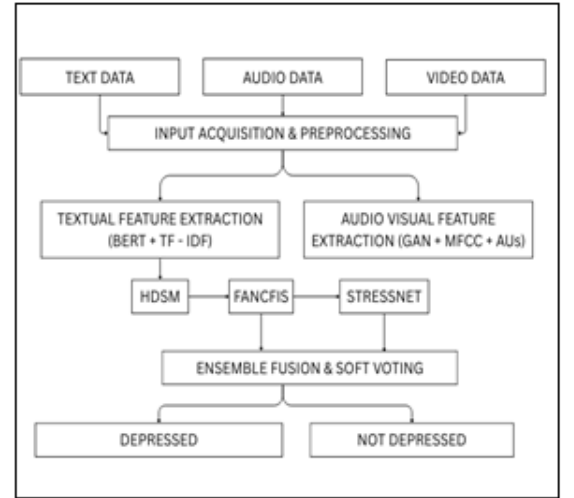


Fig. 1 Proposed Architecture

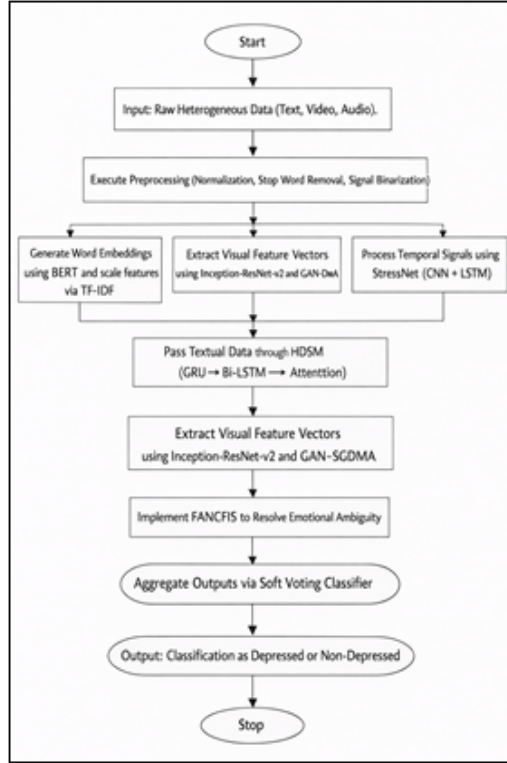


Fig. 2 Proposed Framework

A. Post-Set Identification and Preprocessing

Initially, a post-set $\{D_i, C_i\}$ is identified, where $I = 1, 2, \dots, n$ represents the dataset size, and each entry D_i is labelled with a classification $C_i \in \{\text{Yes}, \text{no}\}$ to denote the stress or depressive state of the user.

1) Text Preprocessing:

Standardization of the input is carried out using normalization, stop words and stemming to remove unnecessary characters and stop words not conveying meaningful information. Word embedding is done through the BERT model. BERT is short for Bidirectional Encoding of Representations from Transformers.

2) Image Preprocessing:

Non-textual information is converted from a raw recording into a digital signal in one dimension. Image processing involves conversion to a grayscale representation to simplify processing, as well as segmentation to align features.

B. Textual Analysis Module (HDSM and FANCFIS)

This approach follows a hierarchical one in analyzing the linguistic pattern that will help in identifying whether the sentiments can be classified as suicidal or non-suicidal.

1) Hybrid Deep Sequential Model (HDSM):

Features are scaled using TF-IDF to quantify the importance of terms. It is then followed by a three-layer

sequential architecture: Layer 1-GRU (Gated Recurrent Unit) by means of unidirectional manner to catch temporal dependencies, and Layer 2-Bi-LSTM network to exploit both past and future contexts of the input sequence. Finally, an attention mechanism was incorporated into it that allows selecting the most informative linguistic segments.

2) Fuzzy Logic Integration:

To deal with the natural vagueness found in human emotive expressions, FANCFIS (Fast Adaptive Neuro Complex Fuzzy Inference System) is implemented. FANCFIS uses features such as the HMLSTM (Hierarchical Multi-scale) approach to learn the lower-level as well as the higher-level features in the sentiment.

C. Audiovisual and Image Modules (GAN-SGDMA and Stress Net)

The behavioural markers are inferred based on non-verbal features to generate a richer construct for mental state assessment.

1) Visual Feature Extraction:

Seven types of facial expression are present, corresponding to various types of emotion, which are extracted using GAN (Generative Adversarial Networks) [3]. In addition, feature vectors are autonomously extracted using the Inception Resnet V2 Model from the video frame and optimal weights are improved by SGDMA (Stochastic Gradient Descent with Momentum Algorithm).

2) Audio and Temporal Dynamics:

The architecture of Stress Net, incorporating 2D-CNN and LSTM, was used to address the temporal dynamics of behavioural signals. Acoustic features were filtered through MFCC (Mel Frequency Cepstral Coefficients) to detect discrepancies in audio and visual streams.

D. Ensemble Fusion Layer

The last stage in the classification is implemented using the modality outcomes in decision level fusion.

1. **Ensemble Learning:** Several base-level classifiers, such as SVM (Support Vector Machine) and Random Forest, are combined in a single model to enhance accuracy over individual models.
2. **Soft Voting Classifier:** In the design of the system, the soft voting classifier is used for aggregating the predictions of the different parts of the system that include the textual part, the visual part, and the audio part on the basis of the cumulative probability scores generated by the different parts of the system. In this way, the system can handle noisy information while relying on high-confidence predictions for diagnosis.

IV. IMPLEMENTATIONS AND RESULTS

To evaluate the performance of the proposed framework, a set of experiments were conducted on a system with a size 16 GB RAM and a CPU of 14 cores. Simulations were conducted by using the Python environment for the Python 3.7 programming language with the addition of the Keras and TensorFlow libraries. In the case of signal processing within the clinical environment, simulation was conducted using the MATLAB 2021a programming environment for the Wavelet Scattering Transform (WST).

A. Dataset Description and Preparation

The datasets used for evaluation are summarized in Table II. Ensure the generalizability of the deep learning model, both the benchmark linguistic databases and the clinical electrophysiological data were used:

TABLE I. DATASET DESCRIPTION AND CHARACTERISTICS

Dataset	Modality	No. of Samples	Classes
SST-2	Text	67,349	Positive / Negative
Sentiment140	Text	1.6 Million	Positive / Negative
Amazon Electronics Reviews	Text	1.68 Million	Positive / Negative
GMC Dataset	EEG	30	Depressed / Healthy
MODMA	Multimodal (EEG, Audio, Video)	53	Depressed / Healthy

- 1) Benchmark Sentiment Corpora: For phrase-level binary sentiment analysis, the Stanford Sentiment Treebank2 (SST2) dataset was used as a benchmark. For large-scale testing, the Sentiment140 dataset, having 1.6 million labelled tweets and the Electronics dataset consisting of 1.68 million Amazon product reviews were used [1].
- 2) Clinical EEG Datasets: GMC provided EEG record data from 30 patients, consisting of 15 healthy and 15 depressed patients, while the Multi-modal Open Dataset for Mental-disorder Analysis (MODMA) had 53 participants in total. [4]
- 3) Preprocessing and Partitioning: Standardized preprocessed steps such as normalization, stop word removal, and stemming were taken to remove noise from the input text [1]. The data is split into a ratio of 80% and 20% for the purposes of training and validation, respectively [8]. Alternatively, a fixed split of 75% for training and 25% for testing has also been taken [1].

B. Simulation Parameters and Optimization

The technical parameters were constantly fine-tuned while trying to optimize the combination of precision and simplicity [8].

1) Optimization Layer:

In this layer, Adam optimizer was used for controlling the rate of convergence, and the initial learning rate was

assigned 0.0001 for better stability in convergence performance [1] [2].

2) Training Schedule:

When training the given model, different epochs were established, ranging from 40 to 200 [1]. As a preventive measure against overfitting of the data within the training period, the inclusion of early stopping with a patience of 10 epochs has been integrated into the model [6].

3) Feature Extraction Settings:

An optimizer was applied to train a network by fine-tuning the inception resnet v2 vision module of 2000 steps with a learning rate of 0.0001 [3]. The scale of WST invariance was chosen to be half the signal's length [4]. Experimental settings and hyperparameters are detailed in Table III.

TABLE II. SIMULATION ENVIRONMENT AND HYPERPARAMETER SETTINGS

Parameter	Value
Programming Language	Python 3.7
Deep Learning Libraries	TensorFlow, Keras
Signal Processing Tool	MATLAB 2021a
Optimizer	Adam
Learning Rate	0.0001
Epochs	40 – 200
Early Stopping	Patience = 10
Train/Test Split	80/20 and 75/25
Hardware	16 GB RAM, 14-Core CPU

C. Performance Evaluation Metrics

Standard indicators were used to quantify the effectiveness of models like Accuracy, Precision, Recall, F1-score, and the Kappa coefficient. Accuracy was evaluated as the ratio of correct predictions to total predictions. Precision and Recall were used for measuring the capability of the classifier to model psychological health and correctly find the positive instances. The Kappa value was estimated to find the agreement between the predicted labels and ground truth, especially for the classification of MDD Architecture-wise and modality-wise and performance comparisons are presented in Table IV and Fig.3.

TABLE III. ARCHITECTURE-WISE PERFORMANCE COMPARISON

Model Architecture	Methodology	Accuracy (%)	Precision (%)	F1-Score (%)
STFS_TF-IDF_HDSM	Hybrid Deep Sequential	98.46	96.71	93.00
FANCFIS	Fuzzy Deep Learning	99.02	98.56	98.45
DWSN-MNN	Wavelet Scattering	99.95	99.95	99.95
Ensemble ML	SVM + RF + Logistic	85.82	86.01	85.81
Baseline CNN	13-Layer CNN	70.32	67.11	77.25

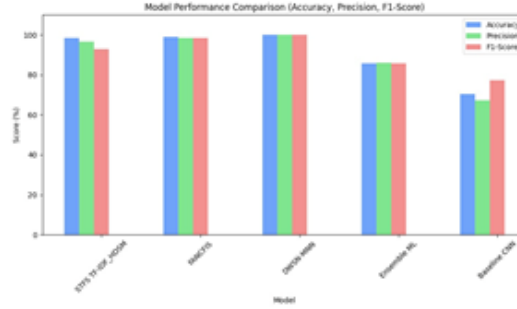


Fig. 3. Architecture-Wise Performance Comparison

D. Quantitative Results and Discussion

TABLE IV. PARAMETERS AND VALUE

Parameter	Value
Optimizer	Adam
Learning Rate	0.0001
Epochs	40-200
Early Stopping	Patience = 10
Train/Test Split	80/20, 75/25

The experimental findings show that multimodal ensemble frameworks achieve significantly higher accuracy than standalone unimodal architectures. As demonstrated in Table IV, the integration of heterogeneous features leads to superior classification stability [2] [10] [4].

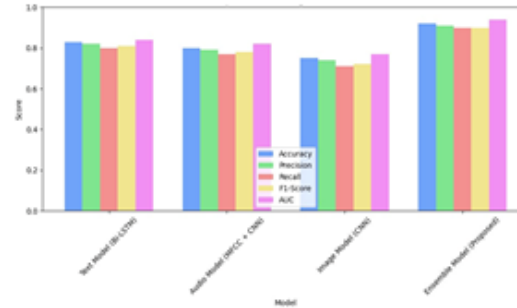


Fig. 4. Modality-Wise Performance Comparison

It was noted that the Hybrid Deep Sequential Model (HDSM) was able to obtain the minimum loss mode of 0.3121. This emphasizes the effectiveness of the model in recognizing the emotional flow using the Bi-Long Short-Term Memory model [5]. In the analysis of the Fast Adaptive Neuro-Complex Fuzzy Inference System (FANCFIS) for the GMC data set, an effective management of the ambiguity involved in the expressions related to sentiments was observed [1]. In the analysis of the right hemisphere EEG signal compared to the left hemisphere for the GMC data set, the higher classification accuracy attained through the right hemisphere emphasizes the involvement of the brain's right hemisphere in the detection of depression

[4]. Accordingly, these analysis results emphasize the effectiveness of the proposed approach involving the use of the Cross-Modal Synergy for the improvement of precision in the professional environment through the achievement of higher AUC values up to 1 for the Scattering Networks [4].

V. CONCLUSION

The present research successfully proves the efficiency of the proposed framework in the context of early depression screening for working professionals. The proposed architecture successfully reduces the subjectivity and scalability limitations of traditional screening processes by utilizing heterogeneous data sources such as text data, audiovisual data, and electrophysiological data. The experimental results confirm the robustness of the cross-modal interaction effect. The Hybrid Deep Sequential Model and its components show high classification accuracy up to 98.46%, while the Deep Wavelet Scattering Networks show a peak classification accuracy of 99.95%. The results reinforce the effectiveness of the proposed framework in combining semantic scaling (TF-IDF), sequential deep learning models (GRU and Bi-LSTM with attention), fuzzy inference systems, and decision-level fusion. As illustrated in the modality-wise comparison in Table VI, the proposed ensemble model outperforms the individual unimodal models in terms of accuracy, precision, recall, and AUC values. Nonetheless, challenges are noted with respect to computational complexity, real-time applicability, and model interpretability. Future research directions should be focused on including more physiological biomarkers, as well as Explainable AI techniques for enhanced model trust from a clinical point of view. The model can be further improved with respect to the cross-attention mechanisms between modalities. In conclusion, it can be noted that the proposed framework for cross-modal depression screening has brought a new dimension with respect to objective and automated methods for evaluating mental health. The proposed system has high potential with respect to early risk detection and assistance from healthcare practitioners.

REFERENCES

- [1] P. Bharath and D. Venkatalakshmi, "FANCFIS: Ensemble deep learning based feature learning with a novel fuzzy approach for sentiment analysis," *International Journal of Information Technology*, vol. 16, pp. 1-12, doi: 10.1007/s41870-024-01882-2, May 2024.
- [2] A. Khan and R. Gupta, "Early depression detection using ensemble machine learning framework," *International Journal of Information Technology*, vol. 16, pp. 1-13, doi: 10.1007/s41870-024-01981-0, Jun. 2024.
- [3] A. K. Cherian, M. Vaidhehi, M. Arshay, J. Briskilal, and S. V. Simpson, "Generative adversarial networks with stochastic gradient descent with momentum algorithm for video-based facial expression," *International Journal of Information Technology*, doi: 10.1007/s41870-024-02043-1, Jun. 2024.
- [4] M. Mahajan et al., "ECG signal classification via ensemble learning: Addressing intra and inter-patient variations," *International Journal of Information Technology*, vol. 16, pp. 4931-4939, doi: 10.1007/s41870-024-02086-4, Aug. 2024.
- [5] F. R. Kareem and A. J. Obaid, "STFS-TF-IDF-HDSM: Semantic and textual feature scaling using TF-IDF based sentiment analysis with Hybrid Deep Sequential Model," *International Journal of Information Technology*, vol. 17, no. 9, pp. 5751-5757, doi: 10.1007/s41870-024-02330-1, Dec. 2025.
- [6] N. Sharma, M. Sharma, J. Tailor, A. Chaudhari, D. Joshi, and U. R. Acharya, "Automated detection of depression using wavelet

- scattering networks," *Medical Engineering and Physics*, vol. 124, Art. no. 104107, doi: 10.1016/j.medengphy.2024.104107, Jan. 2024.
- [7] M. Ying, X. Shao, J. Zhu, Q. Zhao, X. Li, and B. Hu, "EDT: An EEG-based attention model for feature learning and depression recognition," *Biomedical Signal Processing and Control*, vol. 93, Art. no. 106182, doi: 10.1016/j.bspc.2024.106182, Mar. 2024.
- [8] S. A. M. Mane and A. Shinde, "StressNet: Hybrid model of LSTM and CNN for stress detection from electroencephalogram signal (EEG)," *Results in Control and Optimization*, vol. 11, Art. no. 100231, doi: 10.1016/j.rico.2023.100231, May 2023.
- [9] S. Yasin, A. Othmani, I. Raza, and S. A. Hussain, "Machine learning based approaches for clinical and non-clinical depression recognition and depression relapse prediction using audiovisual and EEG modalities: A comprehensive review," *Computers in Biology and Medicine*, vol. 159, Art. no. 106741, doi: 10.1016/j.compbiomed.2023.106741, Mar. 2023.
- [10] V. Vajrobol, N. Aggarwal, U. Shukla, G. J. Saxena, S. Singh, and A. Pundir, "Explainable cross-lingual depression identification based on multi-head attention networks in Thai context," *International Journal of Information Technology*, vol. 15, doi: 10.1007/s41870-023-01512-3, Dec. 2023.

8.6 Copyright Receipt

22/4/26, 7:44 PM

Copyright Office



भारत सरकार / GOVERNMENT OF INDIA

कॉपीराइट कार्यालय / Copyright Office

सौदिक संयबा भवन, लाट संख्या 32, सेक्टर 14, हराका, गई दिल्ली-110078 पहेन: 011-28032496

Intellectual Property Bhawan, Plot No. 32, Sector 14, Dwarka, New Delhi-110078 Phone: 011-28032406

रसीद / Receipt



PAGE No : 1

To,

Shwet Tiwari

B-766 CD height, Yashwant Viva Township, near the Capital Mall, Vasal East, 401209.

(M)-7666453540 : E-mail- Shwettiwari49@gmail.com

RECEIPT NO : 765432

FILING DATE : 22/04/2026

BRANCH : Delhi

USER : Shwet49

S.No	Form	Diary No.	Request No	Title	Amount (Rupees)
1	Form-XIV	SW-17885/2026-CO	269203	SAHAJ - Multimodal Depression Detection	500
Amount in Words		Rupees Five Hundreds			500

PAYMENT MODE	Transaction Id	CIN
Online	C-0000396907	1904260011485

(Administrative Officer)

*This is a computer generated receipt, hence no signature required.

*Please provide your email id with every form or document submitted to the Copyright office so that you may also receive acknowledgements and other documents by email.

8.7 Github Link of Project

<https://github.com/ShwetTiwari/SAHAI-WEB>

8.8 Picture with Guide

